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# Practical 3: Multiple Linear Regression

# 1. Import required libraries
import numpy as np
import pandas as pd

from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# 2. Load the dataset
diabetes = load_diabetes()

X = diabetes.data
y = diabetes.target

# 3. Display dataset
df = pd.DataFrame(X, columns=diabetes.feature_names)
df["Target"] = y

print("First five rows:")
print(df.head())

# 4. Split the dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 5. Create and train the Multiple Linear Regression model
model = LinearRegression()
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model.fit(X_train, y_train)

# 6. Predict the output
y_pred = model.predict(X_test)

# 7. Calculate evaluation metrics
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

# 8. Display results
print("\nMultiple Linear Regression Results")
print("MSE:", mse)
print("RMSE:", rmse)
print("R2 Score:", r2)

# 9. Compare actual and predicted values
result = pd.DataFrame({
    "Actual Values": y_test,
    "Predicted Values": y_pred
})

print("\nActual vs Predicted Values:")
print(result.head(10))
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